TT-P-320D
October 31, 1977
SUPERSEDING
Fed. Spec. TT-P-320C
September 27, 1968

FEDERAL SPECIFICATION

PIGMENT, ALUMINUM: POWDER AND PASTE FOR PAINT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a leafing aluminum pigment for use in protective coatings.

1.2 Classification.

1.2.1 Types and classes. Aluminum pigment covered by this specification shall be of the following types and classes (see 6.1).

Type I - Aluminum pigment powder.

Class A - Extra fine lining.
Class B - Standard lining.
Class C - Standard fineness.

Type II - Aluminum pigment paste.

Class A - Extra fine lining.
Class B - Standard lining.
Class C - Standard fineness.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-V-81 - Varnish, Mixing, for Aluminum Paints.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking Of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 480 - Sampling and Testing Aluminum Powder and Paste.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Type I, aluminum pigment powder. Aluminum pigment powder shall consist of commercially pure aluminum in the form of fine, polished flakes, and a suitable fatty lubricant. It shall contain no filler or extender pigments and shall meet the requirements of table I.

3.2 Type II, aluminum pigment paste. Aluminum pigment paste shall consist of commercially pure aluminum in the form of fine, polished flakes, and a suitable fatty lubricant combined with a volatile thinner to form a paste. It shall contain no filler or extender pigments and shall meet the requirements in table I.

TABLE I. Quantitative requirements

	Type I (Powder)	Type II (Paste)
Nonvolatile matter at 105 to 110°C, min, percent by weight	99	65
Easily extracted fatty and oily matter (polishing lubricant), max, percent by weight	4.0	3.0
Total impurities other than fatty and oily matter, max, percent by weight	1.0	0.7
Coarse particles, max, percent by weight		
Class A (total residue retained on a No. 325 (45-um) sieve	0.1	0.1
Class B (total residue retained on a No. 325 (45-um) sieve	1.5	1.0
Class C (total residue retained on:		
No. 100 (150-um) sieve	0.5	0.5
No. 325 (45-um) sieve	20.0	15.0
Leafing, min, percent:		
Class A	50	55
Class B	50	50
Class C	40	40
Lead, percent by weight of nonvolatile matter, max	0.06	0.06

3.3 Qualitative requirements.

3.3.1 Condition in container (type II only). The aluminum pigment paste, when tested as in 4.4.2, shall be free from separation of pigment and liquid, and shall show no hardening or caking.

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